WEAPONS AND TECHNOLOGY

NAVAL COMMUNICATION

BEFORE THE INTRODUCTION OF RADIO, visual signals were the prime means of fleet communication. Sailing-era navies chiefly relied upon flags, although these had many drawbacks. The admiral's flags could only be seen by ships close to him, and had to be passed on by others raising the same signal. Complex instructions could not be easily conveyed, and once in battle, smoke rendered flags invisible.

FLAG SYSTEMS

During the 17th-century Anglo-Dutch Wars, Britain's Royal Navy introduced a system using five large flags to convey 25 possible signals. The number of signals had grown to 45 by the 1690s, but there was no substantial progress until the mid-18th century, when the French introduced number flags. These allowed hundreds of different signals to be conveyed and were belatedly adopted by the Royal Navy in 1799, which was employing a telegraphic system of communication by the early 1800s. Flags representing letters, and also specific messages, permitted almost unlimited communication.



DISTRESS SIGNAL Flare pistols were developed in World War I to be used by ships in distress.



Nelson's signal Admiral Nelson signals "England expects every man to do his duty" aboard Victory before the battle of Trafalgar

CURRENT MERCANTILE CODES In the International Code of Signals each flag stands for a letter as well as a message.



MORSE CODE

The code invented by American Samuel Morse in the 1840s allowed any letter to be represented by two symbols, for example a dot and a dash, in different combinations. Morse was adopted by navies using flashing signal lights for communication and then for wireless telegraphy at the end of the 19th century. The most famous Morse code message is SOS, adopted as an agreed international distress signal in 1908.

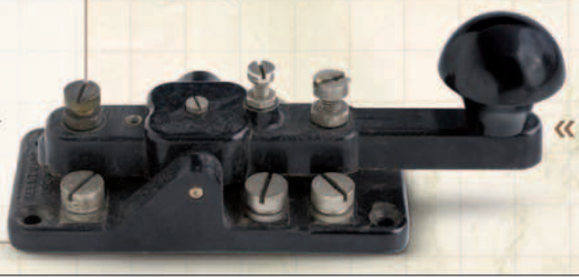
A . –	J . – – –	S . . .
B – . . .	K – . –	T –
C – . – .	L . – . .	U . . –
D – . .	M – –	V . . . –
E .	N – .	W . – –
F . . – .	O – – –	X – . . –
G – – .	P . – – .	Y – . – –
H	Q – – . –	Z – – . .
I . .	R . – .	1 . – – – –

Dots and dashes A "straight key" (below right) is used to transmit the code. The operator controls the timing of the dots and dashes.

1	. – – – –
2	. . – – –
3	. . . – –
4	. . . . –
5	. . . . .
6	– . . . .
7	– – . . .
8	– – – . .
9	– – – – .
0	– – – – –

PENNANTS The tapered type of flag known as a pennant is used primarily to indicate numbers.





STRAIGHT KEY When sending a message, the signal is "on" when the knob is pressed, and "off" when it is released.